

Chapter 2 Limits

2.8 Infinitesimals and Comparison of Infinitesimals

P116a-P116e

- The concept of Infinitesimal
- Properties of Infinitesimals
- Comparison of Infinitesimals
- Substitution by equivalent infinitesimals in computing limits



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SCHOOL OF
MATHEMATICS AND PHYSICS

2.8 Infinitesimals & Comparison of Infinitesimals

无穷小, 无穷小的比较

① The Concept of Infinitesimal

Definition If the **function** $f(x) \rightarrow 0$ as $x \rightarrow c$ (or $x \rightarrow \infty$), then **$f(x)$** is called an **infinitesimal** 无穷小 (infinite small) as $x \rightarrow c$ ($x \rightarrow \infty$).

→ Infinitesimal is NOT a number!

Examples

- x^2 , $\sin x$, $\tan x$, $1 - \cos x$ are all infinitesimals as $x \rightarrow 0$.
- $\sqrt{x} - 1$ is an infinitesimal as $x \rightarrow 1$.
- $a_n = 1/n$ is an infinitesimal as $n \rightarrow \infty$.

Caution Any nonzero constant, no matter how small it is, say, 10^{-100} , **CANNOT** be regarded as an infinitesimal.

Caution $\sin x$ is **NOT** an infinitesimal; $\sin x$ **is** an infinitesimal as $x \rightarrow 0$.



② Properties of Infinitesimals

Theorem A $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} f(x) = L$ if and only if $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} (f(x) - L) = 0$,
that is to say, $f(x) - L$ is an infinitesimal as $x \rightarrow c$ ($x \rightarrow \infty$).

Example We know $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. So $\frac{\sin x}{x} - 1$ is an infinitesimal as $x \rightarrow 0$.

Theorem C (1) If $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} |f(x)| = \infty$, then $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} 1/f(x) = 0$,
that is to say, $1/f(x)$ is an infinitesimal as $x \rightarrow c$ ($x \rightarrow \infty$).
(2) If $f(x)$ is an infinitesimal as $x \rightarrow c$ ($x \rightarrow \infty$) and $f(x) \neq 0$,
then $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} 1/|f(x)| = \infty$.



② Properties of Infinitesimals

Theorem B (1) The sum of finite number of infinitesimals is an infinitesimal.
(2) The product of finite number of infinitesimals is an infinitesimal.
(3) ~~*~~ The product of an infinitesimal and a bounded function is an infinitesimal.
无穷小 乘 有界函数 仍是无穷小.

Example 1 Find $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$.

Solution Notice that $\sin \frac{1}{x}$ is a bounded function since $\left| \sin \frac{1}{x} \right| \leq 1$ ($x \neq 0$), and x is an infinitesimal as $x \rightarrow 0$. So, by the above theorem, we conclude that $x \sin \frac{1}{x}$ is an infinitesimal as $x \rightarrow 0$.

That is $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$.



③ Comparison of Infinitesimals

- Infinitesimals can be classified according to the relative speed with which they approach zero.

➤ It does not make sense to talk about the speed of one infinitesimal, but it does make sense to talk about whether two related infinitesimals approach 0 at a comparable speed.

➤ As $x \rightarrow 0$, both $2x$ and $1000x^2$ are infinitesimals, and $1000x^2$ approaches 0 faster than $2x$.

➤ Even if we multiply by a million, $1,000,000x^2$ still approaches 0 much faster than $2x$ as $x \rightarrow 0$.

x	$2x$	x^2	$1000x^2$
0.1	0.2	0.01	10
0.01	0.02	0.0001	0.1
0.001	0.002	0.000001	0.001
0.0001	0.0002	0.00000001	0.00001
0.00001	0.00002	0.0000000001	0.0000001
0.000001	0.000002	0.000000000001	0.000000001

- We use “^阶order” to formalize this concept of relative speed.



③ Comparison of Infinitesimals

Definition Let $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \alpha(x) = 0$ and $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \beta(x) = 0$.

(1) If $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta(x)}{\alpha(x)} = 0$, we say that $\beta(x)$ is an infinitesimal of **higher order**

than $\alpha(x)$, denoted by $\beta(x) = o(\alpha(x))$ as $x \rightarrow c$ ($x \rightarrow \infty$).

$\beta(x)$ 是 $\alpha(x)$ 的高阶无穷小

(2) If $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta(x)}{\alpha(x)} = 1$, we say that $\beta(x)$ and $\alpha(x)$ are **equivalent**

infinitesimals, denoted by $\beta(x) \sim \alpha(x)$ as $x \rightarrow c$ ($x \rightarrow \infty$).

$\beta(x)$ 和 $\alpha(x)$ 是等价无穷小

Exercise Show that if $\lim_{x \rightarrow c} \frac{\beta(x)}{\alpha(x)} = 1$ then $\lim_{x \rightarrow c} \frac{\alpha(x)}{\beta(x)} = 1$. $\lim_{x \rightarrow 0} \frac{x}{\sin x} = \frac{1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1$



Examples

- It is easy to compute that $\lim_{x \rightarrow 0} \frac{1000x^2}{2x} = 0$, so $1000x^2$ is an infinitesimal of higher order than $2x$, denoted by $1000x^2 = o(2x)$.
- Since $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$, we know $1 - \cos x$ is an infinitesimal of higher order than x , denoted by $1 - \cos x = o(x)$.
- We have found that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2/2} = 1$, so we get the following equivalent infinitesimals as $x \rightarrow 0$:

$$x \sim \sin x \sim \ln(1+x), \quad 1 - \cos x \sim \frac{x^2}{2}.$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \quad \begin{aligned} &\Rightarrow \lim_{x \rightarrow 0} \frac{1 - \cos x}{\frac{1}{2}x^2} = 1 \Rightarrow 1 - \cos x \sim \frac{x^2}{2} \text{ as } x \rightarrow 0. \\ &\Rightarrow \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \cdot x \right) = \frac{1}{2} \cdot 0 = 0. \end{aligned}$$



Example

As $x \rightarrow 0$,

$$x \sim \sin x \sim \ln(1 + x), \quad 1 - \cos x \sim \frac{x^2}{2} = o(x).$$

x	$\sin x$	$\ln(1 + x)$	$x^2/2$	$1 - \cos x$
0.1	$0.9983341664 \times 10^{-1}$	$0.9531017980 \times 10^{-1}$	0.5×10^{-2}	$0.4995834722 \times 10^{-2}$
0.01	$0.9999833334 \times 10^{-2}$	$0.9950330853 \times 10^{-2}$	0.5×10^{-4}	$0.4999958333 \times 10^{-4}$
0.001	$0.9999998333 \times 10^{-3}$	$0.9995003331 \times 10^{-3}$	0.5×10^{-6}	$0.4999999583 \times 10^{-6}$
0.0001	$0.9999999983 \times 10^{-4}$	$0.9999500033 \times 10^{-4}$	0.5×10^{-8}	$0.4999999996 \times 10^{-8}$
0.00001	$0.9999999999 \times 10^{-5}$	$0.9999950000 \times 10^{-5}$	0.5×10^{-10}	$0.4999999999 \times 10^{-10}$
0.000001	$0.9999999999 \times 10^{-6}$	$0.9999994999 \times 10^{-6}$	0.5×10^{-12}	$0.4999999999 \times 10^{-12}$



Examples 2, 4, 5 Show that, as $x \rightarrow 0$,
 (i) $\tan x \sim x$,
 (ii) $\ln(1+x) \sim x$, (iii) $\arcsin x \sim x$, (iv) $e^x - 1 \sim x$.

Solution (i) $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \cdot \frac{1}{\cos x} \right) = 1 \cdot \frac{1}{1} = 1.$

(ii) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) = \lim_{x \rightarrow 0} \ln(1+x)^{1/x} = \ln \left(\lim_{x \rightarrow 0} (1+x)^{1/x} \right) = \ln e = 1.$

(iii) $\lim_{x \rightarrow 0} \frac{\arcsin x}{x} \stackrel{\substack{\theta = \arcsin x \\ x = \sin \theta}}{=} \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = \frac{1}{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}} = \frac{1}{1} = 1.$

(iv) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \stackrel{\substack{u = e^x - 1 \\ x = \ln(1+u)}}{=} \lim_{u \rightarrow 0} \frac{u}{\ln(1+u)} = \frac{1}{\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u}} \stackrel{(ii)}{=} \frac{1}{1} = 1.$

Exercise Show that, as $x \rightarrow 0$,

(i) $\arctan x \sim x$, (ii) $\sqrt{1+x} - 1 \sim \frac{x}{2}$, (iii) $\sqrt[3]{1+x} - 1 \sim \frac{x}{3}$. Rationalization 

Exercise (Paper Assignment, Question 1)

The following functions are all infinitesimals as $x \rightarrow 0$. Indicate the order of each of the following infinitesimals as x to some power n .

For example, the answer to $\sin(x^3)$ is x^3 .

x^n

(1) $x^3 + x^5$, (2) $x \sin x$, (3) $\sqrt{1+x} - \sqrt{1-x}$, (4) $\ln(1+x^2)$.

$\sin(x^3) \sim x^3$. The order is x^3 .

$1 - \cos x \sim \frac{1}{2}x^2$. The order is x^2 .



Important Equivalent Infinitesimals

$$x \sim \sin x \sim \tan x \sim \arcsin x \sim \arctan x \sim (e^x - 1) \sim \ln(1 + x);$$

$$1 - \cos x \sim \frac{x^2}{2};$$

$$(1 + x)^r - 1 \sim rx \ (r \neq 0), \text{ in particular } \sqrt[n]{1 + x} - 1 \sim \frac{1}{n}x.$$

as $x \rightarrow 0$

Example 7' Show that $(1 + x)^n - 1 \sim nx$ as $x \rightarrow 0$ for $n \in \mathbb{N}$.

Recall Binomial Theorem 二项式定理

$$(a + b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2}a^{n-2}b^2 + \cdots + nab^{n-1} + b^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k,$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the **binomial coefficient** 二项式系数.

Proof $(1 + x)^n - 1 = \left(1 + nx + \frac{n(n-1)}{2}x^2 + \cdots + nx^{n-1} + x^n \right) - 1.$

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{nx} = \lim_{x \rightarrow 0} \frac{nx + \text{terms with degree} \geq 2}{nx} = 1.$$



Exercise (Example 7) Show that $(1+x)^r - 1 \sim rx$ as $x \rightarrow 0$ for any r .

Hint Write $(1+x)^r - 1 = e^{\ln(1+x)^r} - 1 = e^{r \ln(1+x)} - 1$.

Solution

$$\left. \begin{array}{l} e^x - 1 \sim x \implies e^{r \ln(1+x)} - 1 \sim r \ln(1+x). \\ \ln(1+x) \sim x \implies r \ln(1+x) \sim rx. \\ \text{So } (1+x)^r - 1 = e^{r \ln(1+x)} - 1 \sim r \ln(1+x) \sim rx. \end{array} \right\} \text{ as } x \rightarrow 0.$$


④ Substitution by Equivalent Infinitesimals in Computing Limits

Theorem D Suppose that $\alpha_1(x), \alpha_2(x), \beta_1(x), \beta_2(x)$ are all infinitesimals as $x \rightarrow c$ ($x \rightarrow \infty$), and $\alpha_1 \sim \alpha_2, \beta_1 \sim \beta_2$; and suppose that

$$\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_2(x)}{\alpha_2(x)} \text{ exists. Then } \lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_1(x)}{\alpha_1(x)} = \lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_2(x)}{\alpha_2(x)}.$$

Example $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{\sin 5x/5x \cdot 5x}{\sin 3x/3x \cdot 3x} = \lim_{x \rightarrow 0} \frac{5x}{3x} = \frac{5}{3}.$

Proof of Theorem D

$$\alpha_1 \sim \alpha_2, \beta_1 \sim \beta_2 \Rightarrow \lim_{x \rightarrow c} \frac{\alpha_1}{\alpha_2} = \lim_{x \rightarrow c} \frac{\beta_1}{\beta_2} = 1.$$

$$\lim_{x \rightarrow c} \frac{\beta_1}{\alpha_1} = \lim_{x \rightarrow c} \frac{\frac{\beta_1}{\beta_2}}{\frac{\alpha_1}{\alpha_2}} \cdot \frac{\beta_2}{\alpha_2} = \frac{1}{1} \cdot \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_2} = \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_2}.$$



Example 6' Find the following limits.

(i) $\lim_{x \rightarrow 0} \frac{\tan x}{x^3 + 3x},$

(ii) $\lim_{x \rightarrow 0} \frac{e^x - 1}{\ln(1 + 2x)},$

(iii) $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + x^2} - 1}{x^2 + 2x^3}$

Exercise Find the limits

$\lim_{x \rightarrow 0} \frac{x^3 + 3x}{3x}$ and $\lim_{x \rightarrow 0} \frac{x^3 + 3x}{x^3}.$

Solution (i) Since $\tan x \sim x$, $x^3 + 3x \sim \underline{3x}$ as $x \rightarrow 0$, we have

$$\lim_{x \rightarrow 0} \frac{\tan x}{x^3 + 3x} = \lim_{x \rightarrow 0} \frac{x}{3x} = \frac{1}{3}.$$

无穷小, 留下低阶的

$\lim_{x \rightarrow \infty} \frac{a_n x^n + \dots + a_1 x + a_0}{b_m x^m + \dots + b_1 x + b_0} = \lim_{x \rightarrow \infty} \frac{a_n x^n}{b_m x^m}$ 无穷大留下高阶的

留下更大的(主项) 舍去小的(余项)

(ii) $\lim_{x \rightarrow 0} \frac{e^x - 1}{\ln(1 + 2x)} = \lim_{x \rightarrow 0} \frac{x}{2x} = \frac{1}{2}.$

As $x \rightarrow 0$, $e^x - 1 \sim x$; $\ln(1+x) \sim x \Rightarrow \ln(1+2x) \sim 2x$

(iii) $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + x^2} - 1}{x^2 + 2x^3} = \lim_{x \rightarrow 0} \frac{\frac{1}{3}x^2}{x^2} = \frac{1}{3}.$

As $x \rightarrow 0$, $(1+x)^{\frac{1}{3}} - 1 \sim \frac{1}{3}x \Rightarrow (1+x^2)^{\frac{1}{3}} - 1 \sim \frac{1}{3}x^2;$

$x^2 + 2x^3 \sim x^2$

As $x \rightarrow 0$,

$x \sim \sin x \sim \tan x$

$x \sim \arcsin x \sim \arctan x$

$x \sim (e^x - 1) \sim \ln(1 + x)$

$1 - \cos x \sim x^2/2$

$(1 + x)^r - 1 \sim rx$



Caution!

- If the numerator or denominator is the **sum** of several terms, then in general, any one of the terms CANNOT be replaced by its equivalent infinitesimal.

如果分子或分母为若干项之和, 那么不能对其中的某个加项作等价无穷小的代换.

Such substitution may possibly result in computational errors.

Example 9 Find $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$. tan $\rightarrow$ sin & cos.

Solution

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} &= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x \cdot (\frac{1}{\cos x} - 1)}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x \cdot (1 - \cos x)}{x^3 \cdot \cos x} \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1 - \cos x}{x^2} \cdot \frac{1}{\cos x} = 1 \cdot \frac{1}{2} \cdot \frac{1}{1} = \frac{1}{2}. \end{aligned}$$

$0 \cdot (1 - 1)$
 $0 \cdot 1$
product & division
instead of
sum & difference

Wrong Answer $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} \neq \lim_{x \rightarrow 0} \frac{x - x}{x^3} = 0.$



Review

Infinitesimals

infinitesimal

infinitesimal of higher order

equivalent infinitesimal

main term

error term

- Determine which infinitesimal is of higher order.
- Determine which infinitesimals are equivalent.
- Use the substitution rule of the equivalent infinitesimals to compute limits.
- As $x \rightarrow 0$,
 - $x \sim \sin x \sim \tan x \sim \arcsin x \sim \arctan x \sim (e^x - 1) \sim \ln(1 + x)$;
 - $1 - \cos x \sim \frac{x^2}{2}$;
 - $(1 + x)^{1/n} - 1 \sim \frac{1}{n}x$. $(1+x)^r - 1 \sim rx$.



Reading Material (**not required**)

Landau's Big O and Little o Notations (**not required**)

- We say that $\beta(x)$ is of **the order of** $\alpha(x)$, denoted by $\beta(x) = O(\alpha(x))$ as $x \rightarrow c$, if there exist positive numbers δ and M such that
$$|\beta(x)| \leq M|\alpha(x)| \quad \text{for all } x \text{ with } 0 < |x - c| < \delta.$$
- In particular, as $x \rightarrow c$, if $\alpha(x)$ and $\beta(x)$ are infinitesimals such that
$$\lim_{x \rightarrow c} \left| \frac{\beta(x)}{\alpha(x)} \right| = L < \infty,$$
 then $\beta(x) = O(\alpha(x))$ as $x \rightarrow c$.
- For example, the followings are all true as $x \rightarrow 0$:

$\sin x = O(x),$	$\sin x = O(1),$
$1 - \cos x = O(x),$	$1 - \cos x = o(x),$
$1 - \cos x = O(x^2),$	$1 - \cos x = O(x^2/2).$



Transitivity of Equivalence of Infinitesimals (not required)

- We know $\sin x \sim x$ and $\tan x \sim x$ as $x \rightarrow 0$.

Does $\sin x \sim \tan x$ hold? **Yes.**

Proof $\lim_{x \rightarrow 0} \frac{\sin x}{\tan x} = \lim_{x \rightarrow 0} \frac{\sin x}{\sin x / \cos x} = \lim_{x \rightarrow 0} \cos x = 1.$

Another Proof $\lim_{x \rightarrow 0} \frac{\sin x}{\tan x} = \lim_{x \rightarrow 0} \frac{\sin x/x}{\tan x/x} = \frac{\lim_{x \rightarrow 0} \sin x/x}{\lim_{x \rightarrow 0} \tan x/x} = \frac{1}{1} = 1.$

- In general, if $\alpha_1(x)$, $\alpha_2(x)$, $\alpha_3(x)$ are all infinitesimals as $x \rightarrow c$, and if $\alpha_1(x) \sim \alpha_2(x)$, $\alpha_2(x) \sim \alpha_3(x)$, then $\alpha_1(x) \sim \alpha_3(x)$. **Transitivity**

Proof $\lim_{x \rightarrow 0} \frac{\alpha_1(x)}{\alpha_3(x)} = \lim_{x \rightarrow 0} \frac{\alpha_1(x)/\alpha_2(x)}{\alpha_3(x)/\alpha_2(x)} = \frac{\lim_{x \rightarrow 0} \alpha_1(x)/\alpha_2(x)}{\lim_{x \rightarrow 0} \alpha_3(x)/\alpha_2(x)} = \frac{1}{1} = 1.$ **传递性**



Main Term and Error Term (not required)

Theorem Suppose that $\alpha(x), \beta(x)$ are both infinitesimals as $x \rightarrow c$. Then
 $\alpha(x) \sim \beta(x)$ if and only if $\alpha = \beta + o(\beta)$.

Here β is called the **main term** 主项 of α .

Proof $\Rightarrow$: If $\alpha \sim \beta$ then $\lim_{x \rightarrow c} \frac{\alpha}{\beta} = 1$. Therefore

$$\lim_{x \rightarrow c} \frac{\alpha - \beta}{\beta} = \lim_{x \rightarrow c} \frac{\alpha}{\beta} - \lim_{x \rightarrow c} \frac{\beta}{\beta} = 1 - 1 = 0,$$

i.e., $\alpha - \beta = o(\beta)$. We can say $\alpha = \beta + o(\beta)$.

$\Leftarrow$: If $\alpha = \beta + o(\beta)$, we have

$$\lim_{x \rightarrow c} \frac{\alpha}{\beta} = \lim_{x \rightarrow c} \frac{\beta + o(\beta)}{\beta} = \lim_{x \rightarrow c} \frac{\beta}{\beta} + \lim_{x \rightarrow c} \frac{o(\beta)}{\beta} = 1 + 0 = 1.$$

That is to say, $\alpha \sim \beta$ as $x \rightarrow c$.



§ 9.9 Taylor Series and Taylor's Formulas (**not required in MTH013**)

As $x \rightarrow 0$,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = x + o(x),$$

$$e^x - 1 = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = x + o(x),$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = x + o(x),$$

$$\cos x - 1 = -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots = -\frac{x^2}{2} + o(x^2).$$

As $x \rightarrow 0$,

$$x \sim \sin x \sim (e^x - 1) \sim \ln(1 + x), \quad 1 - \cos x \sim \frac{x^2}{2}.$$



Another Proof of § 2.8 Theorem D

Theorem D Suppose that $\alpha_1(x), \alpha_2(x), \beta_1(x), \beta_2(x)$ are all infinitesimals as $x \rightarrow c$ ($x \rightarrow \infty$), and $\alpha_1 \sim \alpha_2, \beta_1 \sim \beta_2$; and suppose that $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_2(x)}{\alpha_2(x)}$ exists. Then $\lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_1(x)}{\alpha_1(x)} = \lim_{\substack{x \rightarrow c \\ (x \rightarrow \infty)}} \frac{\beta_2(x)}{\alpha_2(x)}$.

Another Proof (not required)

Recall that $\beta_1 \sim \beta_2$ if and only if $\beta_1 = \beta_2 + o(\beta_2)$. So we have

$$\begin{aligned} \lim_{x \rightarrow c} \frac{\beta_1}{\alpha_1} &= \lim_{x \rightarrow c} \frac{\beta_2 + o(\beta_2)}{\alpha_1} = \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_1} \left(1 + \frac{o(\beta_2)}{\beta_2} \right) \\ &= \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_1} \left(1 + \lim_{x \rightarrow c} \frac{o(\beta_2)}{\beta_2} \right) = (1 + 0) \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_1} = \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_1}. \end{aligned}$$

Similarly one can show $\lim_{x \rightarrow c} \frac{\beta_2}{\alpha_1} = \lim_{x \rightarrow c} \frac{\beta_2}{\alpha_2}$.



§ 2.8 Example 9 Revisited (not required)

- The Taylor's Formulas as $x \rightarrow 0$:
Exercise* Find $\lim_{x \rightarrow 0} \frac{e^x - 1 - \ln(1+x)}{x^2}$.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = x + o(x)$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots = x + o(x)$$

- The right way to find the limit in Example 9 using substitution of equivalent infinitesimal:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} &= \lim_{x \rightarrow 0} \frac{\left(x + \frac{x^3}{3} + o(x^3) \right) - \left(x - \frac{x^3}{3!} + o(x^3) \right)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^3}{3} + \frac{x^3}{3!} + o(x^3)}{x^3} = \frac{1}{3} + \frac{1}{6} + 0 = \frac{1}{2}. \end{aligned}$$

